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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024 (Old)

Subject: Mathematics-1 (Calculus & Linear Algebra)

Code: 105102

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct answer the following (Any seven question only): **[14]**

(a) The curvature of a straight line is:

(i) Infinity
(ii) Zero
(iii) One
(iv) Length of the straight line

(b) The surface area of a solid of revolution generated by rotating $y=f(x)$ about the x-axis is given by:

(i) $\int_a^b 2\pi f(x)dx$
(ii) $\int_a^b \pi f(x)^2 dx$
(iii) $\int_a^b 2\pi f(x)\sqrt{(1+f^{prime}(x)^2)}dx$
(iv) $\int_a^b f(x)dx$

(c) Let f be twice differentiable and suppose that $f^{prime}(c)=0$. Which condition indicates that f has a local minimum at $x=c$?

(i) $f^{primeprime}(c) > 0$
(ii) $f^{primeprime}(c) < 0$
(iii) $f^{primeprime}(c) = 0$
(iv) $f^{primeprime}(c)$ does not exist

(d) Evaluate the limit: $\lim_{x\to 0}[1-\cos x] / [x^2]$

(i) 0
(ii) 1
(iii) 1/2
(iv) 2

(e) Parseval's theorem for a Fourier series relates the Fourier coefficients to the energy of the function. Which statement best describes Parseval's theorem?

(i) The sum of the Fourier coefficients equals the average value of f(x).
(ii) The sum of the squares of the Fourier coefficients equals (up to a constant) the integral of $(f(x)^2)$ over one period.
(iii) The sum of the squares of the Fourier coefficients is proportional to the maximum value of f(x).
(iv) Parseval's theorem provides the Fourier coefficients for discontinuous functions.

(f) Which of the following is the correct Taylor series expansion for $\ln(1+x)$ valid for $|x|<1$?

(i) $\ln(1+x)=x-x^2/2+x^3/3-x^4/4dots$
(ii) $\ln(1+x)=x+x^2/2+x^3/3+dots$
(iii) $\ln(1+x)=1-x-x^2/2+dots$
(iv) $\ln(1+x)=x-x^2+x^3-x^4$

(g) Let $f(x,y)=[x+y] / [x-y]$ if $x \neq y$, and 0 if $x=y$. For which points on the line $x=y$ is $f(x,y)$ continuous?

(i) f is continuous at every point on $x=y$.
(ii) f is continuous only at the origin (0,0).
(iii) f is continuous at all points on $x=y$ except when $x \neq 0$.
(iv) f is discontinuous everywhere on $x=y$.

(h) Let $f(x,y)=e^{xy}$. Which of the following best describes the total derivative of f at a point (a, b)?

(i) The total derivative exists if the partial derivatives exist, regardless of continuity.
(ii) The total derivative is given by $df=e^{ab}(b\,dx + a\,dy)$ evaluated at (a, b).
(iii) f is differentiable only if its partial derivatives are continuous.
(iv) The existence of one zero partial derivative implies the total derivative does not exist.

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(i) Let A be a 5×7 matrix with *rank*(A)=4. According to the Rank-Nullity Theorem, what is the dimension (nullity) of the null space?

(i) 1 (iii) 1/4

(ii) 3 (iv) 7

(j) Consider the system $x+y=5$
$2x+2y=10$. Which of the following best describes the solution set of the system?

(i) A unique solution (iii) Infinitely many solutions

(ii) No solution (iv) A solution only exists for $x=y$

Q.2 Solve both questions : **[14]**

(a) Show that the Gamma function satisfies $\Gamma(n+1)=n!$ For positive integers n.

(b) Find the volume of the solid obtained by rotating the curve $y=x^2$ about the x-axis from $x=0$ to $x=a$.

Q.3 Solve both questions : **[14]**

(a) Evaluate the improper integral $\int_0^\infty e^{-ax}\sin bx\sim dx$ where $a>0$ and $b>0$.

(b) Let $f(x)=x(x-3)^2$, $x$inmathbb$R$.
i. Show that f is continuous on [0,3] and differentiable on (0,3).
ii. Verify that $f(0)=f(3)$.
iii. Use Rolle's Theorem to show that there exists at least one c in (0,3) such that $f^{\text{prime}}(c)=0$. Find the value of c.

Q.4 Solve both questions : **[14]**

(a) Evaluate the limit $\lim_{x\to 0}[\tan x-x] / [x^3]$. Provide a detailed step-by-step application of L'Hôpital's rule.

(b) Consider the function $f(x)=x^3-6x^2+9x+5$.
1. Find the critical points of f by solving $f^{\text{prime}}(x)=0$.
2. Determine the nature (local maximum or minimum) of each critical point using the second derivative test.
3. Find the absolute maximum and minimum of f on the closed interval [0, 5].

Q.5 Solve both questions : **[14]**

(a) Consider the series $\Sigma_{n=1}^\infty[(-1)^n\sqrt(n)] / [n+1]$.
(i) Use the Alternating Series Test to show that the series converges.
(ii) Determine whether the series converges absolutely or only conditionally.

(b) Write the Maclaurin series for sin$x$ up to (and including) the x^5 term. Derive the Maclaurin series for ln(1+x) and state its interval of convergence.

Q.6 Solve both questions : **[14]**

(a) Consider the function $f(x)=x$, 0<x< (*that*="" [−π,π]="="="").="" .<="" π="" a="" are="" as="" b_{n}="[2(−1)^{n+1}] / [n]" be="" by="" coefficients="" expanded="" extension="" f(x)="" fourier="" given="" is="" is,="" odd="" of="" onto="" p="" series="" show="" sine="" that="" the="" to="" which="">
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(b) Let $A=\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix}$.

i. Compute the determinant of A to verify whether A is invertible.
ii. Find the inverse of A.
iii. Determine the rank of A and use the Rank-Nullity theorem to find the nullity of the associated linear transformation T:mathbb$R^3\to$mathbbR^3 defined by $T(x)=Ax$.

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Q.7 Solve both questions : **[14]**

(a) Let $M=\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix}$.

i. Show that M is symmetric.

ii. Find the eigenvalues and corresponding eigenvectors of M.

iii. Diagonalize M: that is, express it in the form $M=PDP^{-1}$ where D is diagonal.

(b) Find and classify the critical points of the function $f(x,y)=x^3-3xy$.

Q.8 Solve both questions : **[14]**

(a) Find the maximum and minimum values of $f(x,y)=x+2y$ subject to the constraint $g(x,y)=x^2-9=0$.

(b) Consider the series $\sum_{n=2}^{\infty}[1] / [n(\ln n)^p]$, $p>0$.

(i) Show that the function $f(x)=[1] / [x(\ln x)^p]$, $x$ge2 is positive, continuous, and decreasing.

(ii) Use the Integral Test to determine for which values of p the series converges.

Q.9 Write short notes on any two of the following: **[14]**

(i) Gradient, curl and divergence with applications.

(ii) Evaluation of definite and improper integrals with applications.

(iii) Applications of definite integrals to evaluate surface areas and volumes of revolutions.

(iv) System of linear equations with applications.

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